

Object Oriented Design Quality Metrics

References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [jqassistant](#)
- [notebook walks through examples for integrating various packages with Neo4j](#)
- [OO Design Quality Metrics](#)
- [A Validation of Martin's Metric](#)
- [Neo4j Python Driver](#)

Incoming Dependencies

Incoming dependencies are also denoted as "Fan-in", "Afferent Coupling" or "in-degree". These are the ones that use the listed package.

If these packages get changed, the incoming dependencies might be affected by the change. The more incoming dependencies, the harder it gets to change the code without the need to adapt the dependent code ("rigid code"). Even worse, it might affect the behavior of the dependent code in an unwanted way ("fragile code").

Since Java Packages are organized hierarchically, incoming dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

Table 1a

- Show the top 20 Java Packages with the most incoming dependencies
- Set the "incomingDependencies" properties on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	incomingDependencies	incomingDependenciesWeight	incon
0	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork	unitofwork	11526	38914	
1	axon-messaging-5.0.2	org.axonframework.messaging.core	core	11118	35970	
2	axon-common-5.0.2	org.axonframework.common.configuration	configuration	3366	18028	
3	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling	eventhandling	3202	10798	
4	axon-common-5.0.2	org.axonframework.common.annotation	annotation	2196	2438	
5	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling	commandhandling	1463	5131	
6	axon-common-5.0.2	org.axonframework.common.infra	infra	1377	1666	
7	axon-common-5.0.2	org.axonframework.common	common	895	2129	
8	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling	queryhandling	698	2468	
9	axon-messaging-5.0.2	org.axonframework.messaging.core.annotation	annotation	487	1508	
10	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	token	425	2147	
11	axon-conversion-5.0.2	org.axonframework.conversion	conversion	415	1708	
12	axon-messaging-5.0.2	org.axonframework.messaging.eventstreaming	eventstreaming	293	833	
13	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.eventstore	eventstore	221	718	
14	axon-messaging-5.0.2	org.axonframework.messaging.tracing	tracing	136	478	
15	axon-test-5.0.2	org.axonframework.test.fixture	fixture	132	359	
16	axon-modelling-5.0.2	org.axonframework.modelling	modelling	125	282	
17	axon-modelling-5.0.2	org.axonframework.modelling.entity	entity	115	394	
18	axon-modelling-5.0.2	org.axonframework.modelling.entity.child	child	89	230	
19	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	segmenting	86	662	

Table 1b

- Show the top 20 Java Packages including their sub-packages with the most incoming dependencies
- Set the property "incomingDependenciesIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	incomingDependencies	incomingDependenciesWeight	incon
0	axon-messaging-5.0.2	org.axonframework.messaging.core	core	18039	59506	
1	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork	unitofwork	11278	38018	
2	axon-common-5.0.2	org.axonframework.common	common	7295	23086	
3	axon-common-5.0.2	org.axonframework.common.configuration	configuration	3205	17368	
4	axon-common-5.0.2	org.axonframework.common.annotation	annotation	2195	2437	
5	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling	eventhandling	2081	7484	
6	axon-common-5.0.2	org.axonframework.common.infra	infra	1363	1592	
7	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling	commandhandling	883	3295	
8	axon-messaging-5.0.2	org.axonframework.messaging.core.annotation	annotation	399	1202	
9	axon-conversion-5.0.2	org.axonframework.conversion	conversion	363	1527	
10	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	token	350	1416	
11	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling	queryhandling	279	897	
12	axon-messaging-5.0.2	org.axonframework.messaging.eventstreaming	eventstreaming	241	688	
13	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	processing	217	685	
14	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	streaming	207	655	
15	axon-messaging-5.0.2	org.axonframework.messaging.tracing	tracing	107	356	
16	axonsourcing-5.0.2	org.axonframework.event sourcing.eventstore	eventstore	77	208	
17	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	segmenting	74	623	
18	axon-modelling-5.0.2	org.axonframework.modelling	modelling	74	226	
19	axon-messaging-5.0.2	org.axonframework.messaging.core.conversion	conversion	66	225	

Outgoing Dependencies

Outgoing dependencies are also denoted as "Fan-out", "Efferent Coupling" or "out-degree". These are the ones that are used by the listed package.

Code from other packages and libraries you're depending on (outgoing) might change over time. The more outgoing changes, the more likely and frequently code changes are needed. This involves time and effort which can be reduced by automation of tests and version updates. Automated tests are crucial to reveal updates, that change the behavior of the code unexpectedly ("fragile code"). As soon as more effort is required, keeping up becomes

difficult (“rigid code”). Not being able to use a newer version might not only restrict features, it can get problematic if there are security issues. This might force you to take “fast but ugly” solutions into account which further increases technical dept.

Since Java Packages are organized hierarchically, outgoing dependencies can be count for every package in isolation or by including all of its sub-packages. The latter one is done without top level packages like for example "org" or "org.company" by assuring that only packages are considered that have other packages or types in the same hierarchy level ("siblings").

Table 2a

- Show the top 20 Java Packages with the most outgoing dependencies
- Set the "outgoingDependencies" properties on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	outgoingDependencies	outgoingDependenciesWeight	outgoin
0	axon-messaging-5.0.2	org.axonframework.messaging.core.configuration	configuration	2463	15546	
1	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	pooled	2321	9770	
2	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	event	1282	4029	
3	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.eventstore.jpa	jpa	1263	4885	
4	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.configuration	configuration	1237	4801	
5	axon-messaging-5.0.2	org.axonframework.messaging.core.interception	interception	1084	5770	
6	axon-test-5.0.2	org.axonframework.test.fixture	fixture	972	3461	
7	axon-modelling-5.0.2	org.axonframework.modelling.entity.annotation	annotation	858	3265	
8	axon-modelling-5.0.2	org.axonframework.modelling.entity	entity	800	2887	
9	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	subscribing	749	2715	
10	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.eventstore	eventstore	700	2610	
11	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	autoconfig	658	2833	
12	axon-modelling-5.0.2	org.axonframework.modelling.configuration	configuration	585	2279	
13	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling	queryhandling	533	2104	
14	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.dist...	distributed	513	1412	
15	axon-messaging-5.0.2	org.axonframework.messaging.monitoring.interce...	interception	500	1550	
16	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.inte...	interception	491	1561	
17	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.anno...	annotation	478	1359	
18	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling	eventhandling	471	1197	
19	axon-messaging-5.0.2	org.axonframework.messaging.core.annotation	annotation	464	1548	

Table 2b

- Show the top 20 Java Packages including their sub-packages with the most outgoing dependencies
- Set the property "outgoingDependenciesIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	outgoingDependencies	outgoingDependenciesWeight	outgo
0	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling	eventhandling	671	2124	
1	axon-messaging-5.0.2	org.axonframework.messaging.core	core	402	1431	
2	axon-modelling-5.0.2	org.axonframework.modelling	modelling	381	1124	
3	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	processing	345	1265	
4	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing	eventsourcing	329	1064	
5	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling	commandhandling	315	1014	
6	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling	queryhandling	314	1122	
7	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	streaming	274	1006	
8	axon-modelling-5.0.2	org.axonframework.modelling.entity	entity	270	856	
9	axon-server-connector-5.0.2	org.axonframework.axonserver.connector	connector	253	866	
10	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	pooled	231	919	
11	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.eventstore	eventstore	194	599	
12	axon-modelling-5.0.2	org.axonframework.modelling.entity.annotation	annotation	179	681	
13	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork	unitofwork	176	693	
14	axon-messaging-5.0.2	org.axonframework.messaging.core.interception	interception	166	587	
15	axon-test-5.0.2	org.axonframework.test	test	160	479	
16	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	event	142	416	
17	axon-test-5.0.2	org.axonframework.test.fixture	fixture	132	472	
18	axon-messaging-5.0.2	org.axonframework.messaging.core.annotation	annotation	124	337	
19	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	autoconfig	121	437	

Instability

$$Instability = \frac{Outgoing\ Dependencies}{Outgoing\ Dependencies + Incoming\ Dependencies}$$

Instability is expressed as the ratio of the number of outgoing dependencies of a module (i.e., the number of packages that depend on it) to the total number of dependencies (i.e., the sum of incoming and outgoing dependencies).

Small values near zero indicate low *Instability*. With no outgoing but some incoming dependencies the *Instability* is zero which is denoted as maximally stable. Such code units are more rigid and difficult to change without impacting other parts of the system. If they are changed less because of that, they are considered stable.

Conversely, high values approaching one indicate high *Instability*. With some outgoing dependencies but no incoming ones the *Instability* is denoted as maximally unstable. Such code units are easier to change without affecting other modules, making them more flexible and less prone to cascading changes throughout the system. If they are changed more often because of that, they are considered unstable.

Since Java Packages are organized hierarchically, *Instability* can be calculated for every package in isolation or by including all of its sub-packages.

Table 3a

- Show the top 20 Java Packages with the lowest *Instability*
- Set the property "instability" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	instability	instabilityTypes	instabilityInterfaces	instabilityF
0	axon-common-5.0.2	org.axonframework.common.annotation	annotation	0.001364	0.021429	0.000000	
1	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork	unitofwork	0.009283	0.080460	0.142857	
2	axon-common-5.0.2	org.axonframework.common.infra	infra	0.019231	0.076336	0.090909	
3	axon-messaging-5.0.2	org.axonframework.messaging.core	core	0.029758	0.163934	0.177083	
4	axon-common-5.0.2	org.axonframework.common	common	0.033477	0.056478	0.000000	
5	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	token	0.036281	0.121212	0.200000	
6	axon-conversion-5.0.2	org.axonframework.conversion	conversion	0.048165	0.156250	0.250000	
7	axon-common-5.0.2	org.axonframework.common.configuration	configuration	0.074257	0.262626	0.295775	
8	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling	eventhandling	0.128233	0.216495	0.392157	
9	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling	commandhandling	0.168750	0.263158	0.431818	
10	axon-common-5.0.2	org.axonframework.common.util	util	0.200000	0.153846	0.000000	
11	axon-messaging-5.0.2	org.axonframework.messaging.core.conversion	conversion	0.238636	0.185185	0.571429	
12	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	store	0.263158	0.190476	1.000000	
13	axon-messaging-5.0.2	org.axonframework.messaging.eventstreaming	eventstreaming	0.263819	0.320513	0.461538	
14	axon-messaging-5.0.2	org.axonframework.messaging.monitoring	monitoring	0.278481	0.195122	0.666667	
15	axon-common-5.0.2	org.axonframework.common.lifecycle	lifecycle	0.333333	0.444444	0.000000	
16	axon-messaging-5.0.2	org.axonframework.messaging.tracing	tracing	0.333333	0.320755	0.533333	
17	axon-update-5.0.2	org.axonframework.update.api	api	0.379310	0.384615	0.000000	
18	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	errorhandling	0.380952	0.384615	0.750000	
19	axon-test-5.0.2	org.axonframework.test.server	server	0.400000	0.400000	0.000000	

Table 3b

- Show the top 20 Java Packages including their sub-packages with the lowest *Instability*
- Set the property "instabilityIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	instability	instabilityTypes	instabilityInterfaces	instabilityF
0	axon-common-5.0.2	org.axonframework.common.annotation	annotation	0.000910	0.014493	0.000000	
1	axon-common-5.0.2	org.axonframework.common.infra	infra	0.003655	0.025210	0.000000	
2	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork	unitofwork	0.015366	0.050420	0.500000	
3	axon-common-5.0.2	org.axonframework.common.configuration	configuration	0.021075	0.070175	0.425287	
4	axon-messaging-5.0.2	org.axonframework.messaging.core	core	0.021799	0.185590	0.391304	
5	axon-conversion-5.0.2	org.axonframework.conversion	conversion	0.059585	0.146341	0.500000	
6	axon-common-5.0.2	org.axonframework.common.util	util	0.090909	0.100000	0.000000	
7	axon-messaging-5.0.2	org.axonframework.messaging.core.conversion	conversion	0.131579	0.160000	0.500000	
8	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	token	0.152542	0.328358	0.500000	
9	axon-messaging-5.0.2	org.axonframework.messaging.eventstreaming	eventstreaming	0.157343	0.204082	0.500000	
10	axon-common-5.0.2	org.axonframework.common.lifecycle	lifecycle	0.166667	0.250000	0.000000	
11	axon-messaging-5.0.2	org.axonframework.messaging.tracing	tracing	0.207407	0.300000	0.461538	
12	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	errorhandling	0.230769	0.333333	0.500000	
13	axon-messaging-5.0.2	org.axonframework.messaging.core.annotation	annotation	0.237094	0.307692	0.421053	
14	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling	eventhandling	0.243823	0.515000	0.566265	
15	axon-common-5.0.2	org.axonframework.common.jdbc	jdbc	0.250000	0.266667	0.000000	
16	axon-common-5.0.2	org.axonframework.common.property	property	0.250000	0.400000	0.000000	
17	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.util	util	0.250000	0.333333	0.000000	
18	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling	commandhandling	0.262938	0.533333	0.527273	
19	axon-modelling-5.0.2	org.axonframework.modelling.repository	repository	0.266667	0.333333	0.500000	

Abstractness

$$Abstractness = \frac{\text{abstract classes in category}}{\text{total number of classes in category}}$$

Package *Abstractness* is expressed as the ratio of the number of abstract classes and interfaces to the total number of classes of a package.

Zero *Abstractness* means that there are no abstract types or interfaces in the package. On the other hand, a value of one means that there are only abstract types.

Since Java Packages are organized hierarchically, *Abstractness* can be calculated for every package in isolation or by including all of its sub-packages.

Table 4a

- Show the top 30 packages with the lowest *Abstractness*
- Set the property "abstractness" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes
0	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	autoconfig	0.0	0	35
1	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot	springboot	0.0	0	22
2	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	event	0.0	0	14
3	axon-metrics-micrometer-5.0.2	org.axonframework.extension.metrics.micrometer	micrometer	0.0	0	12
4	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.inte...	interception	0.0	0	8
5	axon-messaging-5.0.2	org.axonframework.messaging.core.timeout	timeout	0.0	0	8
6	axon-update-5.0.2	org.axonframework.update.api	api	0.0	0	6
7	axon-messaging-5.0.2	org.axonframework.messaging.tracing.attributes	attributes	0.0	0	6
8	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.command	command	0.0	0	6
9	axon-messaging-5.0.2	org.axonframework.messaging.monitoring.interce...	interception	0.0	0	5
10	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.in...	interception	0.0	0	5
11	axon-conversion-5.0.2	org.axonframework.conversion.converter	converter	0.0	0	5
12	axon-conversion-5.0.2	org.axonframework.conversion.json	json	0.0	0	5
13	axon-common-5.0.2	org.axonframework.common.lifecycle	lifecycle	0.0	0	5
14	axon-tracing-opentelemetry-5.0.2	org.axonframework.extension.tracing.opentelemetry	opentelemetry	0.0	0	5
15	axon-update-5.0.2	org.axonframework.update.detection	detection	0.0	0	4
16	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	jpa	0.0	0	4
17	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.inte...	interception	0.0	0	3
18	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	inmemory	0.0	0	3
19	axon-messaging-5.0.2	org.axonframework.messaging.core.configuration...	reflection	0.0	0	3
20	axon-test-5.0.2	org.axonframework.test.server	server	0.0	0	2
21	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.actuato...	axonserver	0.0	0	2
22	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	annotation	0.0	0	2
23	axon-messaging-5.0.2	org.axonframework.messaging.core.reflection	reflection	0.0	0	2
24	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork.an...	annotation	0.0	0	2
25	axon-messaging-5.0.2	org.axonframework.messaging.core.configuration	configuration	0.0	0	2
26	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.eventstore.jdbc	jdbc	0.0	0	2
27	axon-update-5.0.2	org.axonframework.update.common	common	0.0	0	1
28	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.util.jpa	jpa	0.0	0	1
29	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.retry	retry	0.0	0	1

Table 4b

- Show the top 30 packages with the highest *Abstractness* and number of Java Types

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes
142	axon-common-5.0.2	org.axonframework.common.function	function	1.000000	2	2
143	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.actuator	actuator	1.000000	1	1
144	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	streaming	1.000000	1	1
145	axon-common-5.0.2	org.axonframework.common.tx	tx	1.000000	1	1
140	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.co...	configuration	0.750000	3	4
141	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.conf...	configuration	0.750000	3	4
139	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.conf...	configuration	0.714286	10	14
138	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.annotation	annotation	0.700000	7	10
137	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork.tr...	transaction	0.600000	3	5
135	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.util	util	0.571429	4	7
136	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.anno...	annotation	0.571429	4	7
134	axon-modelling-5.0.2	org.axonframework.modelling.repository	repository	0.555556	5	9
133	axon-modelling-5.0.2	org.axonframework.modelling.configuration	configuration	0.538462	7	13
128	axon-common-5.0.2	org.axonframework.common.jdbc	jdbc	0.500000	6	12
129	axon-common-5.0.2	org.axonframework.common.annotation	annotation	0.500000	2	4
130	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.conv...	conversion	0.500000	1	2
131	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.gateway	gateway	0.500000	1	2
132	axon-messaging-5.0.2	org.axonframework.messaging.core.conversion	conversion	0.500000	1	2
127	axon-common-5.0.2	org.axonframework.common.configuration	configuration	0.478261	22	46
125	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling	commandhandling	0.473684	9	19
126	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling	eventhandling	0.473684	9	19
124	axon-messaging-5.0.2	org.axonframework.messaging.core.interception....	annotation	0.454545	5	11
123	axon-test-5.0.2	org.axonframework.test.fixture	fixture	0.451613	14	31
122	axon-messaging-5.0.2	org.axonframework.messaging.eventstreaming	eventstreaming	0.437500	7	16
120	axon-modelling-5.0.2	org.axonframework.modelling.entity.child	child	0.428571	6	14
121	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing	eventsourcing	0.428571	3	7
119	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling	queryhandling	0.416667	10	24
116	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.di...	distributed	0.400000	4	10
117	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.repl...	annotation	0.400000	4	10
118	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.gateway	gateway	0.400000	2	5

Table 4c

- Show the top 30 packages including their sub-packages with the highest package depth and lowest *Abstractness*
- Set the property "abstractnessIncludingSubpackages" on Package nodes.

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSub
0	axon-metrics-micrometer-5.0.2	org.axonframework.extension.metrics.micrometer	micrometer	0.0	0	13	
1	axon-messaging-5.0.2	org.axonframework.messaging.core.configuration	configuration	0.0	0	5	
2	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.autoconfig	autoconfig	0.0	0	35	
3	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.event	event	0.0	0	14	
4	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.inte...	interception	0.0	0	8	
5	axon-messaging-5.0.2	org.axonframework.messaging.core.timeout	timeout	0.0	0	8	
6	axon-update-5.0.2	org.axonframework.update.api	api	0.0	0	6	
7	axon-messaging-5.0.2	org.axonframework.messaging.tracing.attributes	attributes	0.0	0	6	
8	axon-server-connector-5.0.2	org.axonframework.axonserver.connector.command	command	0.0	0	6	
9	axon-messaging-5.0.2	org.axonframework.messaging.monitoring.interce...	interception	0.0	0	5	
10	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.in...	interception	0.0	0	5	
11	axon-conversion-5.0.2	org.axonframework.conversion.converter	converter	0.0	0	5	
12	axon-conversion-5.0.2	org.axonframework.conversion.json	json	0.0	0	5	
13	axon-common-5.0.2	org.axonframework.common.lifecycle	lifecycle	0.0	0	5	
14	axon-tracing-opentelemetry-5.0.2	org.axonframework.extension.tracing.opentelemetry	opentelemetry	0.0	0	5	
15	axon-update-5.0.2	org.axonframework.update.detection	detection	0.0	0	4	
16	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	jpa	0.0	0	4	
17	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.inte...	interception	0.0	0	3	
18	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	inmemory	0.0	0	3	
19	axon-messaging-5.0.2	org.axonframework.messaging.core.configuration...	reflection	0.0	0	3	
20	axon-test-5.0.2	org.axonframework.test.server	server	0.0	0	2	
21	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.actuato...	axonserver	0.0	0	2	
22	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	annotation	0.0	0	2	
23	axon-messaging-5.0.2	org.axonframework.messaging.core.reflection	reflection	0.0	0	2	
24	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork.an...	annotation	0.0	0	2	
25	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.eventstore.jdbc	jdbc	0.0	0	2	
26	axon-update-5.0.2	org.axonframework.update.common	common	0.0	0	1	
27	axon-spring-boot-	org.axonframework.extension.springboot.util.jpa	jpa	0.0	0	1	

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxSub
	autoconfigure-5.0.2						
28	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.retry	retry	0.0	0	1	
29	axon-metrics-micrometer-5.0.2	org.axonframework.extension.metrics.micrometer...	reservoir	0.0	0	1	

Table 4d

- Show the top 30 packages including their sub-packages with the highest package depth and highest *Abstractness*

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	max5
115	axon-common-5.0.2	org.axonframework.common.function	function	1.000000	2	2	
116	axon-common-5.0.2	org.axonframework.common.tx	tx	1.000000	1	1	
113	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.co...	configuration	0.750000	3	4	
114	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.conf...	configuration	0.750000	3	4	
112	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.conf...	configuration	0.714286	10	14	
111	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork.tr...	transaction	0.600000	3	5	
110	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.anno...	annotation	0.571429	4	7	
109	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.annotation	annotation	0.562500	9	16	
108	axon-modelling-5.0.2	org.axonframework.modelling.repository	repository	0.555556	5	9	
107	axon-modelling-5.0.2	org.axonframework.modelling.configuration	configuration	0.538462	7	13	
101	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.util	util	0.500000	4	8	
102	axon-common-5.0.2	org.axonframework.common.jdbc	jdbc	0.500000	6	12	
103	axon-common-5.0.2	org.axonframework.common.annotation	annotation	0.500000	2	4	
104	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.conv...	conversion	0.500000	1	2	
105	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.gateway	gateway	0.500000	1	2	
106	axon-messaging-5.0.2	org.axonframework.messaging.core.conversion	conversion	0.500000	1	2	
100	axon-common-5.0.2	org.axonframework.common.configuration	configuration	0.478261	22	46	
99	axon-messaging-5.0.2	org.axonframework.messaging.core.interception....	annotation	0.454545	5	11	
98	axon-test-5.0.2	org.axonframework.test.fixture	fixture	0.451613	14	31	
97	axon-messaging-5.0.2	org.axonframework.messaging.eventstreaming	eventstreaming	0.437500	7	16	
96	axon-modelling-5.0.2	org.axonframework.modelling.entity.child	child	0.428571	6	14	
93	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.di...	distributed	0.400000	4	10	
94	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.repl...	annotation	0.400000	4	10	
95	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.gateway	gateway	0.400000	2	5	
92	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling	queryhandling	0.380952	24	63	
91	axon-modelling-5.0.2	org.axonframework.modelling	modelling	0.376344	35	93	
89	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling	commandhandling	0.375000	24	64	

	artifactName	fullQualifiedPackageName	packageName	abstractness	numberAbstractTypes	numberTypes	maxS
90	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.ga...	gateway	0.375000	3	8	
88	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.anno...	annotation	0.368421	7	19	
86	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork	unitofwork	0.363636	8	22	

Distance from the main sequence

The *main sequence* is a imaginary line that represents a good compromise between *Abstractness* and *Instability*. A high distance to this line may indicate problems. For example is very *stable* (rigid) code with low abstractness hard to change.

Read more details on that in [OO Design Quality Metrics](#) and [Calculate metrics](#).

Table 5a

- Show the top 30 packages with the highest distance from the "main sequence"

	artifactName	fullQualifiedName	name	distance	abstractness	instability	elementsCount
0	axon-messaging-5.0.2	org.axonframework.messaging.tracing.attributes	attributes	NaN	0.0	NaN	6
1	axon-conversion-5.0.2	org.axonframework.conversion.converter	converter	NaN	0.0	NaN	5
2	axon-messaging-5.0.2	org.axonframework.messaging.core.reflection	reflection	NaN	0.0	NaN	2
3	axon-common-5.0.2	org.axonframework.common.function	function	NaN	1.0	NaN	2
4	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.actuator	actuator	NaN	1.0	NaN	1
5	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.retry	retry	NaN	0.0	NaN	1
6	axon-metrics-micrometer-5.0.2	org.axonframework.extension.metrics.micrometer...	reservoir	NaN	0.0	NaN	1
7	axon-common-5.0.2	org.axonframework.common.io	io	NaN	0.0	NaN	1
8	axon-common-5.0.2	org.axonframework.common.digest	digest	NaN	0.0	NaN	1
9	axon-test-5.0.2	org	org	NaN	0.0	NaN	0
10	axon-test-5.0.2	org.axonframework	axonframework	NaN	0.0	NaN	0
11	axon-update-5.0.2	org	org	NaN	0.0	NaN	0
12	axon-update-5.0.2	org.axonframework	axonframework	NaN	0.0	NaN	0
13	axon-spring-boot-autoconfigure-5.0.2	org	org	NaN	0.0	NaN	0
14	axon-spring-boot-autoconfigure-5.0.2	org.axonframework	axonframework	NaN	0.0	NaN	0
15	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension	extension	NaN	0.0	NaN	0
16	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.service	service	NaN	0.0	NaN	0
17	axon-messaging-5.0.2	org	org	NaN	0.0	NaN	0
18	axon-messaging-5.0.2	org.axonframework	axonframework	NaN	0.0	NaN	0
19	axon-messaging-5.0.2	org.axonframework.messaging	messaging	NaN	0.0	NaN	0
20	axon-conversion-5.0.2	org	org	NaN	0.0	NaN	0
21	axon-conversion-5.0.2	org.axonframework	axonframework	NaN	0.0	NaN	0
22	axon-metrics-micrometer-5.0.2	org	org	NaN	0.0	NaN	0
23	axon-metrics-micrometer-5.0.2	org.axonframework	axonframework	NaN	0.0	NaN	0
24	axon-metrics-micrometer-5.0.2	org.axonframework.extension	extension	NaN	0.0	NaN	0
25	axon-metrics-micrometer-5.0.2	org.axonframework.extension.metrics	metrics	NaN	0.0	NaN	0
26	axon-modelling-5.0.2	org	org	NaN	0.0	NaN	0
27	axon-modelling-5.0.2	org.axonframework	axonframework	NaN	0.0	NaN	0
28	axon-eventsourcing-5.0.2	org	org	NaN	0.0	NaN	0
29	axon-eventsourcing-5.0.2	org.axonframework	axonframework	NaN	0.0	NaN	0

Table 5b

- Show the top 30 packages including their sub-packages with the highest distance from the "main sequence"

	artifactName	fullQualifiedName	name	distance	abstractness	instability	elementsCount
0	axon-common-5.0.2	org.axonframework.common.lifecycle	lifecycle	0.833333	0.000000	0.166667	5
1	axon-conversion-5.0.2	org.axonframework.conversion	conversion	0.773748	0.166667	0.059585	30
2	axon-common-5.0.2	org.axonframework.common.infra	infra	0.746345	0.250000	0.003655	8
3	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	token	0.718425	0.129032	0.152542	31
4	axon-messaging-5.0.2	org.axonframework.messaging.core	core	0.693797	0.284404	0.021799	218
5	axon-update-5.0.2	org.axonframework.update.api	api	0.684211	0.000000	0.315789	6
6	axon-messaging-5.0.2	org.axonframework.messaging.commandhandling.co...	configuration	0.675000	0.750000	0.925000	4
7	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.conf...	configuration	0.675000	0.750000	0.925000	4
8	axon-test-5.0.2	org.axonframework.test.server	server	0.666667	0.000000	0.333333	2
9	axon-messaging-5.0.2	org.axonframework.messaging.core.unitofwork	unitofwork	0.620998	0.363636	0.015366	22
10	axon-common-5.0.2	org.axonframework.common.util	util	0.575758	0.333333	0.090909	6
11	axon-messaging-5.0.2	org.axonframework.messaging.tracing	tracing	0.529435	0.263158	0.207407	19
12	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.anno...	annotation	0.523810	0.571429	0.952381	7
13	axon-common-5.0.2	org.axonframework.common.configuration	configuration	0.500664	0.478261	0.021075	46
14	axon-update-5.0.2	org.axonframework.update.detection	detection	0.500000	0.000000	0.500000	4
15	axon-update-5.0.2	org.axonframework.update.common	common	0.500000	0.000000	0.500000	1
16	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.util.jpa	jpa	0.500000	0.000000	0.500000	1
17	axon-common-5.0.2	org.axonframework.common.annotation	annotation	0.499090	0.500000	0.000910	4
18	axon-eventsourcing-5.0.2	org.axonframework.eventsourcing.annotation	annotation	0.474781	0.562500	0.912281	16
19	axon-messaging-5.0.2	org.axonframework.messaging.core.annotation	annotation	0.442906	0.320000	0.237094	50
20	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling	eventhandling	0.440388	0.315789	0.243823	171
21	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.proc...	errorhandling	0.435897	0.333333	0.230769	3
22	axon-common-5.0.2	org.axonframework.common.property	property	0.416667	0.333333	0.250000	9
23	axon-modelling-5.0.2	org.axonframework.modelling.configuration	configuration	0.415174	0.538462	0.876712	13
24	axon-messaging-5.0.2	org.axonframework.messaging.eventstreaming	eventstreaming	0.405157	0.437500	0.157343	16
25	axon-spring-boot-autoconfigure-5.0.2	org.axonframework.extension.springboot.actuato...	axonserver	0.400000	0.000000	0.600000	2
26	axon-messaging-5.0.2	org.axonframework.messaging.queryhandling.gateway	gateway	0.400000	0.500000	0.900000	2
27	axon-messaging-5.0.2	org.axonframework.messaging.eventhandling.conf...	configuration	0.371589	0.714286	0.657303	14
28	axon-messaging-5.0.2	org.axonframework.messaging.core.conversion	conversion	0.368421	0.500000	0.131579	2
29	axon-common-5.0.2	org.axonframework.common.jpa	jpa	0.365385	0.250000	0.384615	4

Abstractness vs. Instability Plot with "Main Sequence" line as reference

- Plot *Abstractness* vs. *Instability* of all packages
- Draw the "main sequence" as dashed green diagonal line
- Scale the packages by the number of types they contain
- Color the packages by their distance to the "main sequence" (blue=near, red=far)

Figure 5a - Packages without their sub-packages

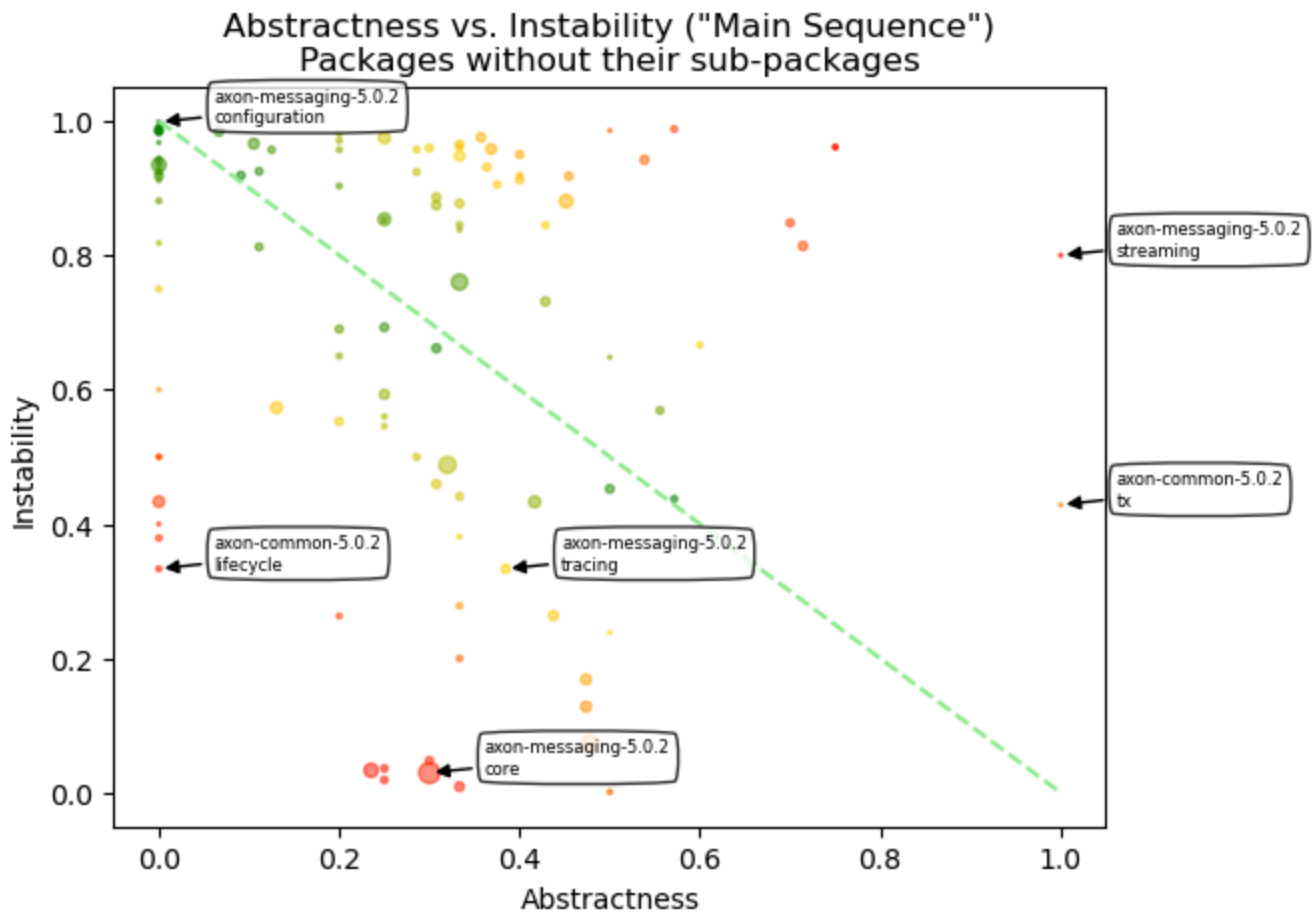


Figure 5b - Packages including their sub-packages

Abstractness vs. Instability ("Main Sequence")
Packages including their sub-packages

